

A Convergent-Sequence Subcase Of A Weak-Star Basic Sequence Problem

FA/Banach smoke run packet

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Source And Target

The source paper is Jerzy Kakol, Manuel Lopez-Pellicer, and Wieslaw Sliwa, *On weak*-basic sequences in duals and biduals of spaces $C(X)$ and Quojunctions*, arXiv:2601.19873.

The target is Problem 12 of the paper:

Does $C_p(X)^*$ admit a weak*-basic sequence for any infinite compact space X ?

The screenshot below is a full-width crop from page 7 of the source PDF.

Theorem 2 may also suggest the following

Problem 12 Does $C_p(X)^*$ admit a weak*-basic sequence for any infinite compact space X ?

Problem 13 Does $C_k(X)^*$ admit a weak*-basic sequence for any infinite Tychonoff space X ?

Candidate Partial Result

Theorem 1. *Let X be a compact Hausdorff space containing a nontrivial convergent sequence. Then $C_p(X)^*$ admits a weak*-basic sequence.*

Proof. Let (x_n) be a sequence of pairwise distinct points of X converging to a point $x \in X \setminus \{x_n : n \in \mathbb{N}\}$. For each n , define

$$\mu_n = \delta_{x_n} - \delta_x \in C_p(X)^*.$$

Here δ_t denotes point evaluation at t . Since the topology of $C_p(X)$ is pointwise convergence, every finite linear combination of point evaluations is continuous.

Let $K = \{x\} \cup \{x_n : n \in \mathbb{N}\}$. This set is compact and closed in X . We first show that

$$M := \text{span}\{\mu_n : n \in \mathbb{N}\}$$

is weak* closed in $C_p(X)^*$, where the weak* topology is $\sigma(C_p(X)^*, C_p(X))$.

Take $\nu \in \overline{M}^{w^*}$. Since the continuous dual of $C_p(X)$ consists of finitely supported signed measures, write

$$\nu = \sum_{j=1}^m a_j \delta_{t_j}.$$

If some $t_j \notin K$ has nonzero coefficient, normality of the compact Hausdorff space X gives a function $f \in C(X)$ such that $f(t_j) = 1$ and f vanishes on K and on the remaining finitely many support points of ν . Every element of M vanishes on this f , so $\nu(f) = 0$, a contradiction. Thus $\text{supp } \nu \subset K$.

The support of ν is finite, hence

$$\nu = a\delta_x + \sum_{n=1}^N b_n\delta_{x_n}$$

for some N . Every $\mu \in M$ annihilates the constant function 1, so the same is true of ν by weak* closure. Therefore $a + \sum_{n=1}^N b_n = 0$, and consequently

$$\nu = \sum_{n=1}^N b_n(\delta_{x_n} - \delta_x) = \sum_{n=1}^N b_n\mu_n \in M.$$

So M is weak* closed.

It remains to check that (μ_n) is a Schauder basis for M with the subspace weak* topology. Each element of M has a unique finite expansion in the sequence (μ_n) . For each fixed n , the point x_n is isolated in the compact set K , so $K \setminus \{x_n\}$ is closed in X . By Urysohn's lemma there is $f_n \in C(X)$ such that

$$f_n(x_n) = 1, \quad f_n|_{K \setminus \{x_n\}} = 0.$$

Then, for every $\sum_k c_k\mu_k \in M$,

$$\left(\sum_k c_k\mu_k \right) (f_n) = c_n.$$

Thus the n th coefficient functional is weak* continuous on M . Since all expansions in M are finite, the partial sums converge trivially in the subspace topology. Hence (μ_n) is a weak*-basic sequence in $C_p(X)^*$. $\square$

Verification Notes

The proof uses only standard facts: the continuous dual of $C_p(X)$ consists of finitely supported point evaluations, compact Hausdorff spaces are normal, and a nontrivial convergent sequence plus its limit is closed compact.

The key verification point is the closed-span argument. The total-mass condition comes from evaluating at the constant function 1, while support outside the convergent compactum is excluded by Urysohn separation.

Limitations And Novelty Check

This is a partial result only. It does not address compact spaces with no nontrivial convergent sequence.

There is also a substantial novelty caveat. The source paper later proves a metrizable-quotient theorem for $C_p(X)$; after observing that restriction to the convergent compact subset is an open quotient onto an infinite-dimensional metrizable locally convex space, this subcase appears to be implicit in the paper's own methods. The value of this packet is therefore the direct constructive proof, not a claim of a full new answer to Problem 12.

Human Review Recommendation

Likely valid as a constructive partial result. Do not count it as a full solve of Problem 12.